

The wireless cocktail party

This is a really interesting way to understand the various methods of channel allocation, called the Cocktail Party Theory. Imagine a large room filled with a variety of people. The TDMA users are those who need to use the specific time slots to say their bits, and they take turns to cycle around their time slots and converse with each other. In FDMA, the users are grouped into widely separate clumps, with each one holding conversations at the same time, but independent of the others. The CDMA folks, on the

other hand, are the people who converse with each other all at the same time. To differentiate between one another, one pair in the cocktail party speaks Tamil, another speaks Hindi, a third speaks French and so on. The Tamil-speaking couple just hones in on the Tamil and ignores everything else as random noise. The Hindi-speaking couple does likewise, and so do the French speakers. The key to CDMA is to extract the desired signal while rejecting everything else as random noise.

the signal performs a time-correlation operation to detect the specific desired codeword. All the unsynchronised signals then appear as background noise. This lets the recipient home in and lock into the desired signal, apply the user's code and recover the message.

CDMA can be implemented in both 2G and 3G networks—the commercial deployment since 1995, was pioneered by Qualcomm, which holds several patents to components of the technology. The 2G implementation of CDMA is called cdmaOne, and is based on the IS-95 standard. There are two modified standards to the IS-95, the IS-95 A and B. They differ in data throughput, with IS-95B coming close to 115 Kbps.

The CDMA architecture

The typical CDMA network architecture consists of four major components, namely the mobile phone or the handheld, the Base Station Transceiver Subsystem (BTS), Mobile Telephone Switching Office (MTSO) and the Public Switched Telephone Network (PSTN). The PSTN provides the access to conventional landline phones located in individual homes and offices. The network topology of CDMA consists of relatively high-powered BTSs that are strategically located and sufficiently dispersed within the region to ensure a proper quality of service. GSM on the other hand, uses lots of relatively low-power transmitters that are clustered together. Here, the same frequencies are reused in neighbouring BTSs, unlike in

vast and infinite resource, with just about everybody laying claim to significant usable chunks of it, in reality, it is a scarce resource. The essential differences in the two systems are in the techniques of allocation and usage of the spectrum.

Spread that spectrum

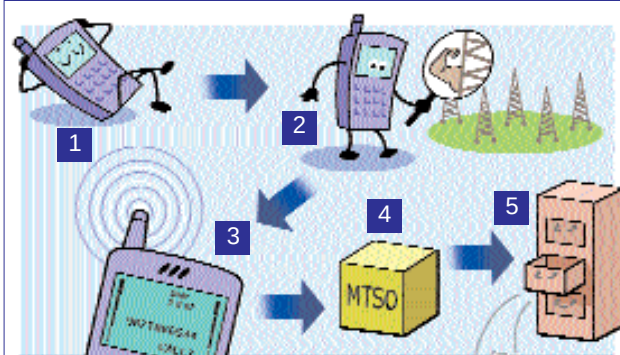
CDMA is a digital wireless technology that makes intelligent use of the radio spectrum. For all mobile wireless services, there is a fine line between the number of users and their total capacity. It is a classical problem—you need sufficient bandwidth to maintain a decent quality of service, but not so much that it becomes vastly unprofitable. In order to make efficient use of the radio spectrum, multiple access technologies were designed that divided the allotted spectrum by time or frequency, thus letting a number of users make use of the communications system. The division of the spectrum on the basis of time is known as TDMA (Time Division Multiple Access). Here each user is allotted a time slot on a specific frequency after the expiry of which, the time slot is then used by another user. TDMA thus tends to work in a round-robin fashion.

The other technique used is allotting frequency-based slots to all the users. This is known as FDMA (Frequency Division Multiple Access). Here each user is allocated a frequency i.e., a radio channel that remains in use till the user has terminated the connection. Once the channel is free, it does not get reallocated or cannot be used by other users to increase capacity. Hence it tends to be a wasted resource. The interesting thing here is that most of the digital wireless technologies depend on either TDMA or

FDMA, or a combination of both, to solve the channel allocation problem. Here is where CDMA comes in—as a channel allocation system. It doesn't divide the spectrum in terms of time or frequency, instead it allows all users on a CDMA network to share and occupy the same frequency, at the same time. CDMA allows each station to transmit over the entire frequency spectrum all the time. The multiple simultaneous transmissions are separated using coding, where each conversation is uniquely coded and can be understood only by the party that is called.

CDMA uses a form of the Spread Spectrum technology. The principle of Spread Spectrum theory is the use of a larger bandwidth carrier wave than is required to transmit the message signals. The carrier waves essentially cloak the message signals and provide the required bandwidth to transmit the message. Here the narrowband message signal is 'spread' over the carrier wave and hence the name. At any given time all the stations use the same carrier frequency to transmit at the same time and are operating independently of each other, with one user having no knowledge of the other. To ensure security, each of the multiple simultaneous conversations are encoded with a chip sequence or code. The intended recipient of

When you Switch on the Phone



- 1 The phone is switched off...
- 2 The phone is turned on and starts scanning for the strongest signal from all BTSs
- 3 Once it catches a strong signal...
- 4 The phone pages the MTSO...
- 5 The MTSO updates the HLR and registers the phone